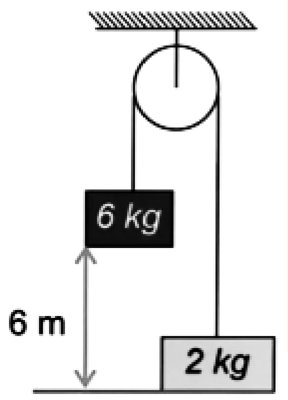
JEE-Main-06-04-2026 (Memory Based)
[EVENING SHIFT]
Physics

Question: Minimum deviation for an equilateral prism is 30°, refractive index is
Options:
(a) √2
(b) √3/2
(c) 2
(d) 4
Answer: (a)

Question: If system given below is released from rest, then find speed of 6 kg block just before hitting ground.
($g = 10\ m/s^2$)



Options:
(a) 6.20 m/s
(b) 7.74 m/s
(c) 4.70 m/s
(d) 5.20 m/s
Answer: (b)

Question: Find electric field, for given electrostatic potential field, at point p(2, 3) ; v = $5(x^2-y^2)$.
Options:
(a) $-20\hat{i} + 30\hat{j}$
(b) $20\hat{i} + 30\hat{j}$

(c) $30\hat{i} - 20\hat{j}$

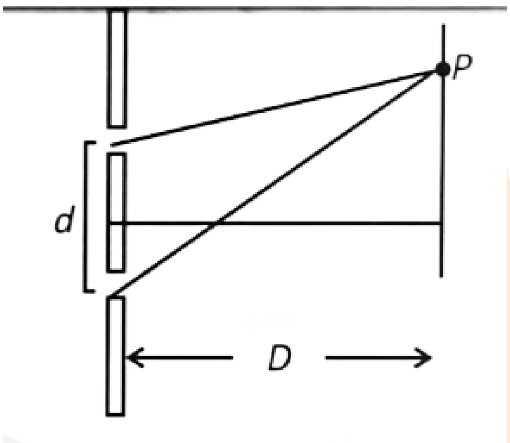
(d) $30\hat{i} + 20\hat{j}$

Answer: (a)

Question: If percentage change in radius of sphere is 2%, then find % change in the volume of sphere.
Answer: (6)

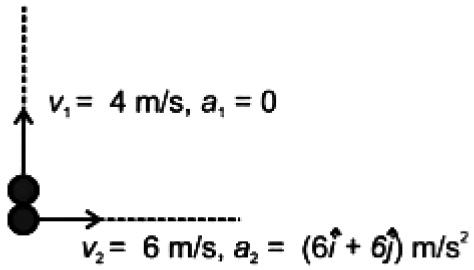
Question: $H_1^2 + H_1^2 \rightarrow He_2^4$
Binding energy per nucleon of H_1^2 and He_2^4 are 1.1 MeV and 7 MeV respectively. Find energy released in the nuclear reaction given above.
Options:
(a) 23.6 MeV
(b) 24.a MeV
(c) 5.9 MeV
(d) 3 MeV
Answer: (a)

Question: In standard YDSE wavelength λ = 7000 Å, d = 5 mm; D = 50 cm . I_0 is the intensity due to individual source on the screen. P is the point on screen such that intensity of 'P' is equal to I_0. Find the minimum distance of 'P from center of screen.



Options:
(a) 35 μm
(b) $\frac{70}{3}\ \mu m$
(c) 17.5 μm
(d) 50 μm
Answer: (b)

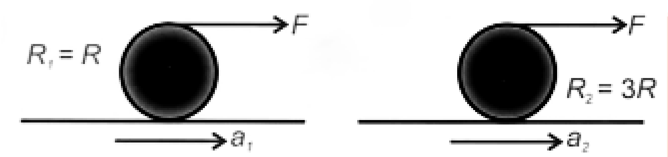
Question: For two equal masses. One of the mass is having initial velocity v_1 = 4 m/s $\hat{j}$ and acceleration a_1= 0 m/s^2. Other having Initial velocity of V_2 = 6 m/s $\hat{i}$ and acceleration = (6 $\hat{i}$ + 6 $\hat{j}$)m/s^2 . Initially both were located at origin.
Find the trajectory of centre of mass.



Options:
(a) Circular
(b) Ellipse
(c) Parabolic
(d) Straight line
Answer: (c)

Question: When mass 200 g hangs from ceiling via spring in equilibrium, the extension in the spring is observed to be 2 mm. Find angular frequency of its SHM & energy stored in spring in equilibrium position are respectively
Options:
(a) 50 rad/s & 2 mJ
(b) $50\sqrt{2}$ rad/s & 2 mJ
(c) 100 rad/s & 4 mJ
(d) 150 rad/s & 4 mJ
Answer: (b)

Question: Solid sphere (1) of mass 5 M, hollow sphere (2) of mass M are pulled tangentially without slipping. Acceleration a1 and a2 are in ratio of _______ . Radius are shown in diagram



Options:
(a) 5/21
(b) 1
(c) 3/7
(d) 5/3
Answer: (a)

Question: 2 moles of ideal mono-atomic gas at temperature T and 6 moles of ideal diatomic gas at temperature 2T are mixed together. Find the equilibrium temperature of mixture.
Options:
(a) $\frac{13}{6}T$
(b) $\frac{11}{8}T$

(c) $\frac{13}{8}T$

(d) $\frac{11}{6}T$

Answer: (d)

Question: n inclined plane is inclined at angle 45° with horizontal. If a block is released an it, it takes twice the time of or it to come down compared to if it was smoot. Find friction coefficient of the inclined plane.

Options:

(a) 1/2

(b) 1/4

(c) 3/4

(d) 1.2

Answer: (c)

Question: Spherical charged body having potential 100 volt and capacitance C_0= 100 μF is brought into contact with identical (uncharged) spherical body. Find the loss of energy.

Options:

(a) 250 mJ

(b) 100 mJ

(c) 50 mJ

(d) 150 mJ

Answer: (a)

Question: Propagation vector $\vec{K}$ & electric field vector $\vec{E}$ of an EMW of angular frequency(ω) is given. Magnetic field will be

Options:

(a) $\frac{\vec{E} \times \vec{K}}{\omega}$

(b) $\left(\vec{E} \times \vec{K}\right)\omega$

(c) $\frac{\vec{K} \times \vec{E}}{\omega}$

(d) $\left(\vec{K} \times \vec{E}\right)\omega$

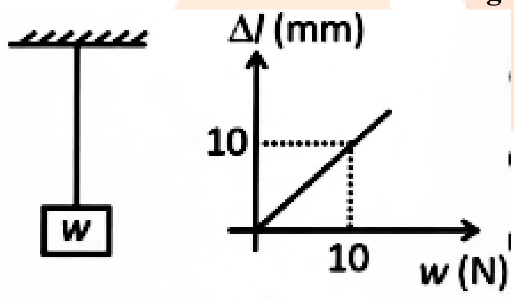
Answer: (c)

Question: Match the following lists and choose correct options:

	List -I		List - II
A	h (Planck's constant)	1	$ML^oT^{-3}K^{-4}$
B	G (Gravitational constant)	2	$M^1L^2T^{-1}$
C	σ (Stefans constant)	3	$M^1L^2T^{-2}k^{-1}$
D	k (Boltzmann constant)	4	$M^{-1}L^3T^{-2}$

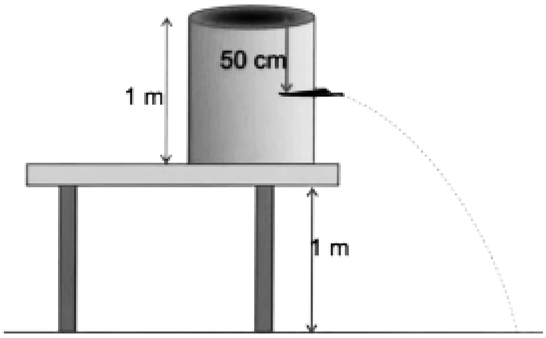
Options:
(a) A→2;B→4;C→1;D→3
(b) A→2;B→4;C→2;D→1
(c) A→2;B→1;C→4;D→3
(d) A→1;B→2;C→3;D→4
Answer: (a)

Question: Graph is between weight of b block & extension in wire. Area of cross-section of wire is 10^{-5} m^2 & length 1 m. Find its Youngs modulus.



Options:
(a) 10^{11} N/m^2
(b) 10^8 N/m^2
(c) 10^{12} N/m^2
(d) 10^{10} N/m^2
Answer: (b)

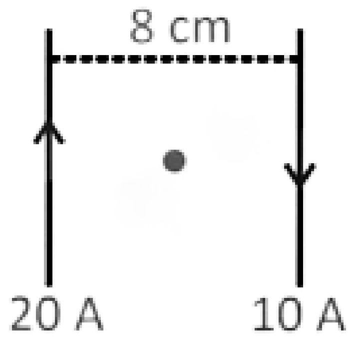
Question: In diagram shown below, hole is made in completely filled cylinder 50 cm below the top. Find range covered by leaking water on the ground.



Options:
(a) $\sqrt{3}$ m
(b) $\sqrt{2}$ m

(c) √4 m
(d) √6 m
Answer: (a)

Question: Magnetic field at middle of two parallel wires carrying current of 20 A and 10 A in opposite direction is ______ μT.



Answer: (150)

JEE-Main-06-04-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: Highest X-O and bond order is, (X is central atom)
Options:
(a) F_2O
(b) CO
(c) H_2O
(d) CO_2
Answer: (b)
Solution: (b) CO

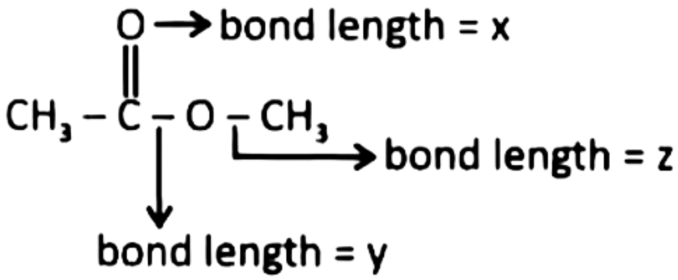
Question: The hybridisation states of Ni in the 3 complexes $Ni(CO)_4$, $[Ni(NH_3)_6]^{2+}$, & $[Ni(CN)_4]^{2-}$ are,
Options:
(a) dsp^2, sp^3d_2, sp^3
(b) sp^3, sp^3d^2, sp^3
(c) sp^3, sp^3d^2, dsp^2
(d) sp^3, d^2sp^3, dsp^2
Answer: (c)
Solution: (c) sp^3, sp^3d^2, dsp^2

Question: In a period, ionisation, energy of the extreme left ad electronegativity of extreme right element is respectively____. (Don't consider Noble gases).
Options:
(a) Lowest/Highest
(b) Lowest/Lowest
(c) Highest/Lowest
(d) Highest/Highest
Answer: (a)
Solution: (a) Lowest/Highest

Question: Given,
$$k = Ae^{\frac{2800K}{T}}$$
Find activation energy
Options:
(a) 23.28 kJ/mol
(b) 56 kcal/mol
(c) 232.8 kJ/mol

(d) 5600 kcal/mol
Answer: (a)
Solution: (a) 23.28 kJ/mol

Question: Compare the bond length x, y and z in the following compound



Options:
(a) x = y = z
(b) x = y < z
(c) x < y < z
(d) x < y = z
Answer: (c)
Solution: (c) x < y < z

Question: Energy of hydrogen like species is given as 54.4 eV. The n and z respectively are
Options:
(a) 1, 2
(b) 2, 2
(c) 2, 1
(d) 1, 1
Answer: (a)
Solution: (a) 1, 2

Question: Match: for ideal monoatomic gas,

A) Isothermal reversible expansion	**1)** $w = -nRT ln\frac{V_f}{V_i}$
B) Adiabatic reversible expansion	**2)** $w = \frac{nR}{\gamma-1}\left(T_f - T_i\right)$
C) Adiabatic irreversible expansion	**3)** $w = nC_v\left(T'_f - T_i\right)$
D) Free expansion	**4) w = 0**

Options:
(a) A-4, B-2, C-3, D-1
(b) A-2, B-1, C-3, D-4
(c) A-1, B-2, C-3, D-4
(d) A-1, B-3, C-2, D-4
Answer: (c)

Solution: (c) A-1, B-2, C-3, D-4

Question: Consider the following statements and choose the correct option:

Statement I: Out of SF_4, SF_6, H_2S, SO_2 and SO_3 molecules, only 4 molecules do not follow octet rule

Statement II: H_2O, SO_2 and H_2S have only 1 lone pair on central atom

Options:

(a) Statement I is correct, statement II is incorrect

(b) Statement I is incorrect, statement II is correct

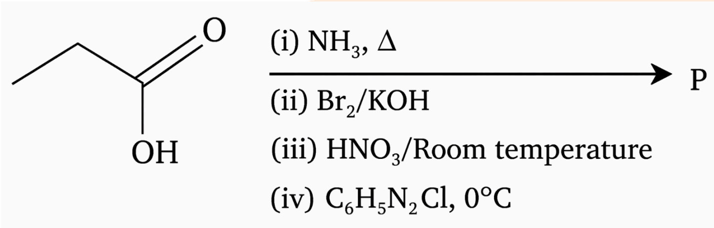
(c) Both statement I and statement II are correct

(d) Both statement I and statement II are incorrect

Answer: (a)

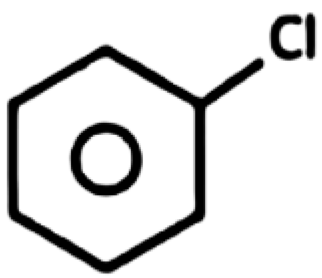
Solution: (a) Statement I is correct, statement II is incorrect

Question: Consider the reaction,

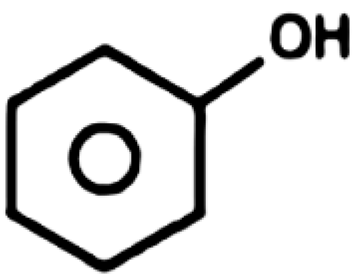


P is

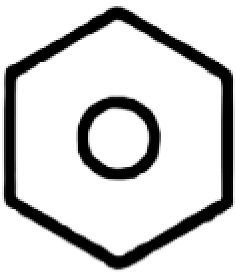
Options:



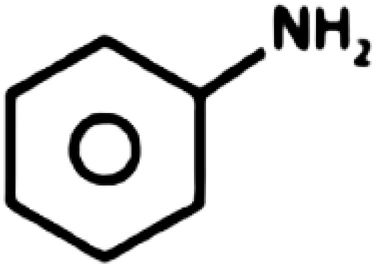
(a)



(b)

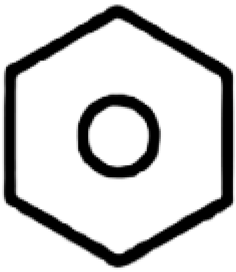


(c)



(d)

Answer: (c)



Solution: (c)

Question: Consider the following reaction

$A \xrightarrow[+conc.\,H_2SO_4]{FeSO_4(fresh)} B(Brown\,ring)$

'A' and 'B' can be respectively

Options:

(a) $NaNO_2$, $[Fe(H_2O)_5NO]^{2+}$

(b) $NaNO_3$, $[FeCH_2O)_6]^{2+}$

(c) $NaNO_2$, $[Fe(H_2O)_6]^{2+}$

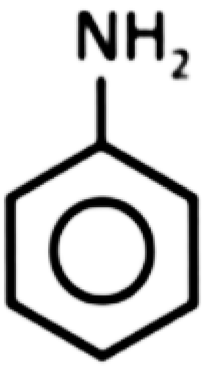
(d) $NaNO_3$, $[Fe(H_2O)_4(NO_2)_2]^{2+}$

Answer: (a)

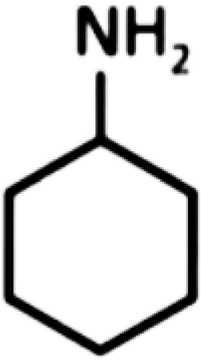
Solution: (a) $NaNO_2$, $[Fe(H_2O)_5NO]^{2+}$

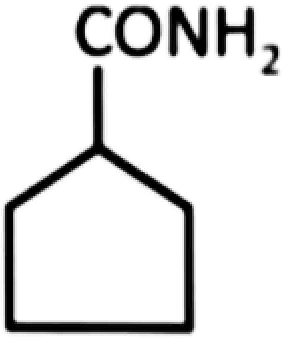
Question: Which of the following can be obtained from phthalimide reaction and also give carbylamine reaction.

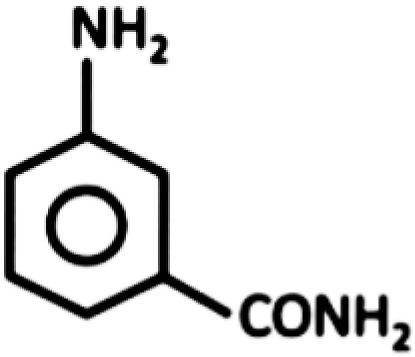
Options:



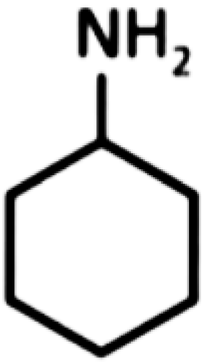
(a)

(b) 

(c) 

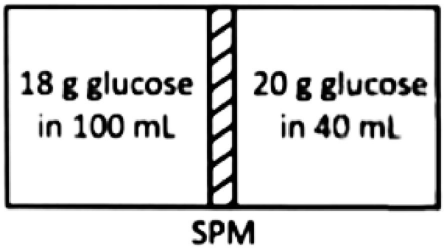
(d) 

Answer: (b)



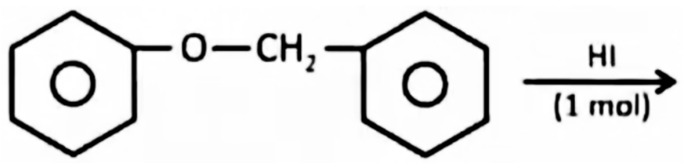
Solution: (b)

Question:



Statement I: Glucose solution moves from vessel II to vessel I through SPM.
Statement II: The osmotic pressure of vessel II is greater than that of vessel I.
Options:
(a) Statement I and statement II both correct
(b) Statement I and statement II both incorrect
(c) Statement I correct and statement II incorrect
(d) Statement I incorrect and statement II correct
Answer: (c)
Solution: (c) Statement I correct and statement II incorrect

Question: Consider the statements in respect of following reaction and choose the correct option.



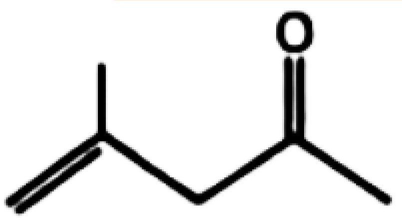
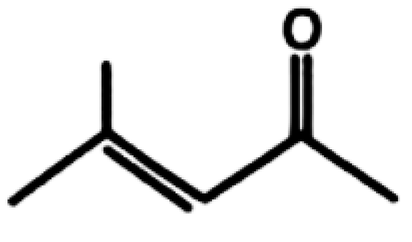
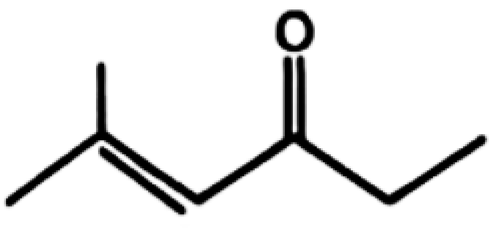
Statement-I: Cleavage of O - CH_2 bond is taking place.
Statement-II: Iodobenzene and benzyl alcohol are product of this reaction.
Options:
(a) Statement-I is correct and Statement-II is incorrect
(b) Statement-I is incorrect and Statement-II is correct
(c) Both of Statement-I and Statement-II are correct
(d) Both of Statement-I and Statement-II are incorrect
Answer: (a)
Solution: (a) Statement-I is correct and Statement-II is incorrect

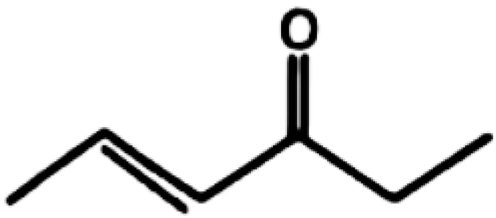
Question: The concentration of A_xB_y is CM and its dissociation constant is k. Find the degree of dissociation (α) fo A_xB_y. Consider α to be negligible in comparison to 1
Options:
(a) $\left(\frac{k}{c^{(x+y-1)} \cdot x^x \cdot y^y}\right)^{\frac{1}{(x+y)}}$
(b) $\left(kc^{(x+y-1)} \cdot x^x y^y\right)^{(x+y)}$
(c) $\left(\frac{k}{c^{(x+y-1)} x^x y^y}\right)^{(x+y)}$
(d) $\left(c^{(x+y-1)} \cdot x^x \cdot y^y\right)^{\frac{1}{(x+y)}}$
Answer: (a)

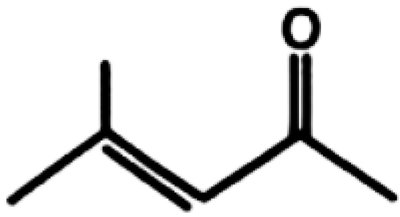
Solution: (a) $\left(\frac{k}{c^{(x+y-1)}\cdot x^{x}\cdot y^{y}}\right)^{\frac{1}{(x+y)}}$

Question: Choose the correct option.
Options:
(a) IE_1 of Cr > IE_1 of Mn
(b) IE_2 of Cr > IE_2 of Mn
(c) IE_2 of Cr < IE_1 of Mn
(d) IE_2 of Mn > IE_2 of Cr
Answer: (b)
Solution: (b) IE_2 of Cr > IE_2 of Mn

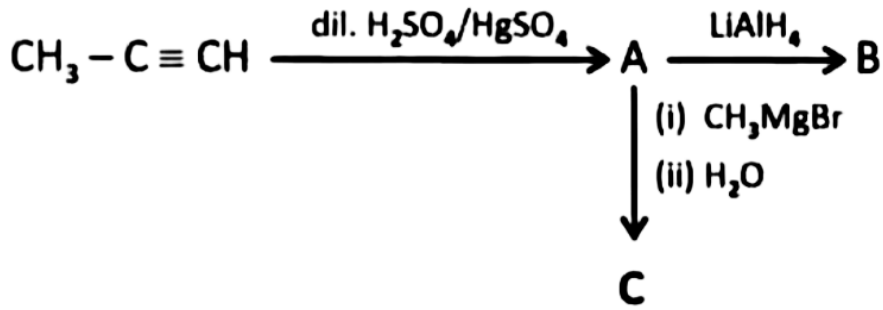
Question: For the reaction, $C_4H_8(g) \rightleftharpoons 2C_2H_4(g)$ 75% dissociation of C_4H_8 is observed and total equilibrium pressure is atm at 298 K.
Find ΔG° in KJ/mol (nearest integer)
$\left(log\frac{9}{1.75} = 0.71\right)$.
R = 8.3 KJ^{-1} mol^{-1}
Options:
Answer: (4)
Solution:

Question:
$CH_3CN \xrightarrow[(ii)H_3O^+]{(i)CH_3MgBr} P \quad CH_3C{\equiv}CH \xrightarrow[H_2SO_4]{Hg^{2+}} Q \quad P + Q \xrightarrow[dil.\Delta]{Ba(OH)_2} Product$
Options:

(a)

(b)

(c)

(d) 
Answer: (b)

Solution: (b) 

Question: Consider the following reaction sequence



Which of the following test can be used to distinguish between B and C
Options:
(a) Lucas test
(b) Fehling solution
(c) Haloform test
(d) Tollen's test
Answer: (c)
Solution: (c) Haloform test

JEE-Main-06-04-2026 (Memory Based)
[EVENING SHIFT]
Maths

Question: The area (in square units) of the region $\{(x, y) : x^2 - 8x \le y \le -x\}$ is

Options:

(a) $\frac{343}{6}$

(b) $\frac{241}{6}$

(c) $\frac{221}{6}$

(d) $\frac{323}{6}$

Answer: (a)

Solution:

Step 1: Find the Points of Intersection

Set the two equations equal to each other to find where the boundaries meet:

$$x^2 - 8x = -x$$

$$x^2 - 7x = 0$$

$$x(x - 7) = 0$$

The points of intersection occur at $x = 0$ and $x = 7$.

Step 2: Set Up the Integral

The area A is the integral of the upper curve ($y = -x$) minus the lower curve ($y = x^2 - 8x$) from $x = 0$ to $x = 7$:

$$A = \int_0^7 [(-x) - (x^2 - 8x)]\, dx$$

$$A = \int_0^7 (7x - x^2)\, dx$$

Step 3: Evaluate the Integral

Now, we find the antiderivative and apply the limits:

$$A = \left[\frac{7x^2}{2} - \frac{x^3}{3}\right]_0^7$$

$$A = \left(\frac{7(7^2)}{2} - \frac{7^3}{3}\right) - (0)$$

$$A = \frac{343}{2} - \frac{343}{3}$$

To subtract these, we find a common denominator (6):

$$A = \frac{3 \cdot 343 - 2 \cdot 343}{6}$$

$$A = \frac{343(3-2)}{6}$$

$$A = \frac{343}{6}$$

Question: Sum of $1+\frac{1}{2}\left(1^2+2^2\right)+\frac{1}{3}\left(1^2+2^2+3^2\right)+\ldots.$ **upto terms is**

Options:

(a) $\frac{313}{2}$

(b) $\frac{315}{2}$

(c) 313

(d) 315

Answer: (b)

Solution:

Step 1: Find the general term (T_k)

The k-th term of the series is given by:

$$T_k = \frac{1}{k}\left(1^2+2^2+3^2+\ldots+k^2\right)$$

We know the formula for the sum of the squares of the first k natural numbers is:

$$\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$$

Substituting this into our expression for T_k:

$$T_k = \frac{1}{k} \cdot \frac{k(k+1)(2k+1)}{6} = \frac{(k+1)(2k+1)}{6}$$

$$T_k = \frac{2k^2+3k+1}{6}$$

Step 2: Set up the sum for 10 terms

We want to find $S_{10} = \sum_{k=1}^{10} T_k$:

$$S_{10} = \sum_{k=1}^{10} \frac{2k^2+3k+1}{6}$$

$$S_{10} = \frac{1}{6}\left[2\sum_{k=1}^{10} k^2 + 3\sum_{k=1}^{10} k + \sum_{k=1}^{10} 1\right]$$

Step 3: Evaluate the sum

Using the standard summation formulas for $n = 10$:

- $\sum_{k=1}^{10} k^2 = \frac{10(11)(21)}{6} = 385$

- $\sum_{k=1}^{10} k = \frac{10(11)}{2} = 55$
- $\sum_{k=1}^{10} 1 = 10$

Now, substitute these values back into the equation:

$$S_{10} = \frac{1}{6}[2(385) + 3(55) + 10]$$

$$S_{10} = \frac{1}{6}[770 + 165 + 10]$$

$$S_{10} = \frac{945}{6}$$

Dividing both the numerator and denominator by 3:

$$S_{10} = \frac{315}{2}$$

Question: $\int_{-1}^{1} \frac{x^3 + |x| + 1}{x^2 + |x| + 1} dx$

Options:

(a) $\ln 3 - \frac{\pi}{3\sqrt{3}}$

(b) $\ln 3 + \frac{\pi}{3\sqrt{3}}$

(c) $\frac{\pi}{3\sqrt{3}}$

(d) $\frac{-\pi}{3\sqrt{3}}$

Answer: (b)

Solution:

Step 1: Simplify using Even and Odd Properties

Let's split the integrand into two parts:

$$f(x) = \underbrace{\frac{x^3}{x^2 + |x| + 1}}_{\text{Part 1}} + \underbrace{\frac{|x| + 1}{x^2 + |x| + 1}}_{\text{Part 2}}$$

- **Part 1 is an odd function:** Replacing x with $-x$ makes the numerator negative while the denominator remains the same. The integral of an odd function over a symmetric interval $[-1, 1]$ is always **0**.
- **Part 2 is an even function:** Replacing x with $-x$ leaves the function unchanged. For even functions, we can double the integral over the interval $[0, 1]$.

So, the integral simplifies to:

$$I = 0 + 2\int_0^1 \frac{x+1}{x^2 + x + 1} dx$$

Step 2: Evaluate the Simplified Integral

To integrate $\frac{x+1}{x^2+x+1}$, we rewrite the numerator in terms of the derivative of the denominator $(2x+1)$:

$$x+1=\frac{1}{2}(2x+1)+\frac{1}{2}$$

Substituting this back into the integral:

$$I=2\int_0^1\left[\frac{\frac{1}{2}(2x+1)}{x^2+x+1}+\frac{\frac{1}{2}}{x^2+x+1}\right]dx$$

$$I=\int_0^1\frac{2x+1}{x^2+x+1}\,dx+\int_0^1\frac{1}{x^2+x+1}\,dx$$

Step 3: Solve the two components

1. The Logarithmic Part:
Using the substitution $u=x^2+x+1$:

$$\int_0^1\frac{2x+1}{x^2+x+1}\,dx=\left[\ln(x^2+x+1)\right]_0^1=\ln(3)-\ln(1)=\ln 3$$

2. The Arctan Part:

Complete the square for the denominator: $x^2+x+1=(x+\frac{1}{2})^2+\frac{3}{4}$.

$$\int_0^1\frac{1}{(x+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\,dx=\left[\frac{1}{\frac{\sqrt{3}}{2}}\tan^{-1}\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\right]_0^1$$

$$=\frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)\right]_0^1=\frac{2}{\sqrt{3}}\left[\tan^{-1}(\sqrt{3})-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right]$$

$$=\frac{2}{\sqrt{3}}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)=\frac{2}{\sqrt{3}}\left(\frac{\pi}{6}\right)=\frac{\pi}{3\sqrt{3}}$$

Question: The value of x for which $\sin^{-1}\left(\frac{2}{3}\sqrt{1-x^2}\right)=\cot^{-1}(2\sqrt{x})$ **is**

Options:

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{8}$

(d) $\frac{1}{9}$

Answer: (a)

Solution:

Step 1: Set up the equation

Let the equation be:

$$\theta = \sin^{-1}\left(\frac{2}{3}\sqrt{1-x^2}\right) = \cot^{-1}(2\sqrt{x})$$

From the right side of the equation:

$$\cot\theta = 2\sqrt{x}$$

Step 2: Use trigonometric identities

We know the identity relating $\sin\theta$ and $\cot\theta$:

$$\sin^2\theta = \frac{1}{1+\cot^2\theta}$$

From the left side of the equation, we have $\sin\theta = \frac{2}{3}\sqrt{1-x^2}$. Squaring both sides gives:

$$\sin^2\theta = \frac{4}{9}(1-x^2)$$

Now, substitute both expressions into the identity:

$$\frac{4}{9}(1-x^2) = \frac{1}{1+(2\sqrt{x})^2}$$

$$\frac{4}{9}(1-x^2) = \frac{1}{1+4x}$$

Step 3: Solve for x

Cross-multiply to remove the fractions:

$$4(1-x^2)(1+4x) = 9$$

$$4(1+4x-x^2-4x^3) = 9$$

$$4+16x-4x^2-16x^3 = 9$$

$$16x^3+4x^2-16x+5 = 0$$

Now, we can test the options provided to see which one satisfies this cubic equation. Let's try **Option A ($x = 1/2$)**:

$$16\left(\frac{1}{2}\right)^3+4\left(\frac{1}{2}\right)^2-16\left(\frac{1}{2}\right)+5$$

$$16\left(\frac{1}{8}\right)+4\left(\frac{1}{4}\right)-8+5$$

$$2+1-8+5=0$$

Since $x = 1/2$ satisfies the equation and falls within the valid domain for the square roots ($x \geq 0$ and $1-x^2 \geq 0$), it is the correct solution.

Question: If matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}$ **and matrix** $[b_{ij}] = B = A^{99} - I_{3\times3}$**, then the value of** $\left(\frac{b_{31} + b_{32}}{b_{21}}\right)$ **is equal to**

Options:

(a) 147

(b) 149

(c) 160

(d) 159

Answer: (b)

Solution:

Step 1: Decompose Matrix A

Notice that matrix A can be written as the sum of the identity matrix I and a strictly lower triangular matrix N:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 3 & 3 & 0 \end{bmatrix} = I + N$$

Step 2: Find Powers of N

Let's calculate N^2 and N^3 to see if the matrix is nilpotent:

- $N^2 = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 3 & 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 3 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 0 & 0 \end{bmatrix}$
- $N^3 = N^2 \cdot N = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 3 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Since $N^3 = 0$, N is nilpotent of order 3. All powers N^k where $k \geq 3$ will also be the zero matrix.

Step 3: Expand A^{99} using the Binomial Theorem

Since I and N commute ($IN = NI$), we can use the binomial expansion for $(I + N)^{99}$:

$$A^{99} = (I + N)^{99} = I + \binom{99}{1} N + \binom{99}{2} N^2 + \ldots$$

Because $N^3 = 0$, the series terminates:

$$A^{99} = I + 99N + \frac{99 \times 98}{2} N^2$$

Now, determine matrix B:

$$B = A^{99} - I = 99N + 4851N^2$$

Step 4: Identify the elements of B

Let's find the specific elements required for the calculation:

- b_{21}: This is the element at row 2, col 1. From $99N$, it is $99 \times 3 = 297$. (Note: N^2 has 0 at this position).
- b_{32}: This is the element at row 3, col 2. From $99N$, it is $99 \times 3 = 297$.
- b_{31}: This is the element at row 3, col 1. It combines values from $99N$ and $4851N^2$:

$$b_{31} = (99 \times 3) + (4851 \times 9)$$

Step 5: Calculate the final value

Substitute these into the given expression:

$$\frac{b_{31} + b_{32}}{b_{21}} = \frac{[(99 \times 3) + (4851 \times 9)] + [99 \times 3]}{99 \times 3}$$

Divide each term in the numerator by the denominator (99×3):

$$\text{Value} = \frac{99 \times 3}{99 \times 3} + \frac{4851 \times 9}{99 \times 3} + \frac{99 \times 3}{99 \times 3}$$

$$\text{Value} = 1 + \frac{4851}{99} \cdot \frac{9}{3} + 1$$

$$\text{Value} = 1 + (49 \times 3) + 1$$

$$\text{Value} = 2 + 147 = \mathbf{149}$$

Question: $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$, $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$ **If** $2\vec{a} + 3\vec{b}$ and $\vec{a} - \vec{b}$ **are two adjacent sides of a triangle, then the square of area of triangle is**

Options:

(a) 1800

(b) 900

(c) 902

(d) 680

Answer: (a)

Solution:

Step 1: Simplify the Side Vectors

Let the two adjacent side vectors be $\vec{u}$ and $\vec{v}$:

- $\vec{u} = 2\vec{a} + 3\vec{b}$
- $\vec{v} = \vec{a} - \vec{b}$

The area of the triangle A is given by:

$$A = \frac{1}{2} |\vec{u} \times \vec{v}|$$

Using the properties of the cross product:

$$\vec{u} \times \vec{v} = (2\vec{a} + 3\vec{b}) \times (\vec{a} - \vec{b})$$

$$\vec{u} \times \vec{v} = 2(\vec{a} \times \vec{a}) - 2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{a}) - 3(\vec{b} \times \vec{b})$$

Since $\vec{a} \times \vec{a} = 0$, $\vec{b} \times \vec{b} = 0$, and $\vec{b} \times \vec{a} = -(\vec{a} \times \vec{b})$:

$$\vec{u} \times \vec{v} = -2(\vec{a} \times \vec{b}) - 3(\vec{a} \times \vec{b}) = -5(\vec{a} \times \vec{b})$$

So, the magnitude is:

$$|\vec{u}\times\vec{v}| = 5|\vec{a}\times\vec{b}|$$

Step 2: Calculate $|\vec{a}\times\vec{b}|$ and $|\vec{a}\times\vec{b}|^2$

Given $\vec{a} = 2\hat{i}+3\hat{j}+3\hat{k}$ and $\vec{b} = 6\hat{i}+3\hat{j}+3\hat{k}$:

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 3 \\ 6 & 3 & 3 \end{vmatrix}$$

$$\vec{a}\times\vec{b} = \hat{i}(9-9) - \hat{j}(6-18) + \hat{k}(6-18)$$

$$\vec{a}\times\vec{b} = 0\hat{i} + 12\hat{j} - 12\hat{k}$$

Now, find the square of the magnitude:

$$|\vec{a}\times\vec{b}|^2 = 0^2 + 12^2 + (-12)^2 = 144 + 144 = \mathbf{288}$$

Step 3: Calculate the Square of the Area (A^2)

The area is $A = \frac{1}{2}(5|\vec{a}\times\vec{b}|)$. Squaring this gives:

$$A^2 = \frac{25}{4}|\vec{a}\times\vec{b}|^2$$

$$A^2 = \frac{25}{4}(288)$$

$$A^2 = 25\times 72$$

$$A^2 = \mathbf{1800}$$

Question: Consider the function f : R → R such that $f(x) = \dfrac{2x^2+x+2}{3x^2+2x+3}$ **then f(x) is**

Options:

(a) One-one & onto

(b) One-one & into

(c) Many one & onto

(d) Many one & into

Answer: (d)

Solution:

1. Check for One-one or Many-one

A function is one-one if every x maps to a unique y. For a rational function with quadratic expressions:

- **Check limits at infinity:**

$$\lim_{x\to\infty} f(x) = \frac{2}{3} \quad \text{and} \quad \lim_{x\to-\infty} f(x) = \frac{2}{3}$$

- **Check continuity:** The denominator $3x^2+2x+3$ has a discriminant $D = 2^2 - 4(3)(3) = -32$. Since $D < 0$ and the leading coefficient is positive, the denominator is always positive and never zero. Thus, $f(x)$ is continuous for all real x.

Since the function is continuous and approaches the same value (2/3) at both positive and negative infinity, it must "turn back" at some point (it has a maximum or minimum). This means it will pass through the same y-values at least twice.

- **Conclusion:** The function is **Many-one**.

2. Check for Onto or Into

A function $f:\ \mathbb{R}\to\mathbb{R}$ is onto if its range is equal to its codomain (all real numbers). Let's find the range by setting $y = f(x)$:

$$y = \frac{2x^2+x+2}{3x^2+2x+3}$$

$$y(3x^2+2x+3) = 2x^2+x+2$$

$$(3y-2)x^2+(2y-1)x+(3y-2) = 0$$

For x to be a real number, the discriminant (D) of this quadratic equation must be ≥ 0:

$$D = (2y-1)^2 - 4(3y-2)(3y-2) \geq 0$$

$$(2y-1)^2 - 4(3y-2)^2 \geq 0$$

Using the identity $a^2 - b^2 = (a-b)(a+b)$:

$$[(2y-1) - 2(3y-2)]\cdot[(2y-1)+2(3y-2)] \geq 0$$

$$(-4y+3)(8y-5) \geq 0$$

Multiply by -1 to simplify:

$$(4y-3)(8y-5) \leq 0$$

This inequality holds when y is between the roots: **$5/8 \leq y \leq 3/4$**.

Since the range $[5/8, 3/4]$ is much smaller than the codomain $\mathbb{R}$, the function does not cover all real numbers.

- **Conclusion:** The function is **Into**.

Final Answer:

The function is **Many-one and into**. The correct option is **D**.

Question: A circle has centre in the 1st quadrant and touches the x-axis at a distance of 3 units from the origin and cutoff an intercept of 6√3 on y-axis. Then, the length of chord having equation x - y = 1 intercepted by the circle is

Options:

(a) $2\sqrt{7}$

(b) $3\sqrt{7}$

(c) $\sqrt{7}$

(d) $4\sqrt{7}$

Answer: (d)

Solution:

Step 1: Find the center and radius of the circle

- The circle touches the x-axis at $(3, 0)$. This means the x-coordinate of the center is $\boldsymbol{h = 3}$.
- Since it touches the x-axis, the radius r is equal to the y-coordinate of the center (k). So, the center is **$(3, r)$**.
- The formula for the y-intercept of a circle is $2\sqrt{k^2-c}$. From the equation $(x-h)^2 + (y-k)^2 = r^2$, we can expand it to find c:

$$(x-3)^2 + (y-r)^2 = r^2 \Rightarrow x^2 - 6x + 9 + y^2 - 2ry + r^2 = r^2$$

$$x^2 + y^2 - 6x - 2ry + 9 = 0$$

Here, the constant term $\boldsymbol{c = 9}$ and the coefficient of y is $2f = -2r$, so $\boldsymbol{f = -r}$.

- Given the y-intercept is $6\sqrt{3}$:

$$2\sqrt{r^2-9} = 6\sqrt{3}$$

$$\sqrt{r^2-9} = 3\sqrt{3}$$

$$r^2 - 9 = 27 \Rightarrow r^2 = 36 \Rightarrow \mathbf{r = 6}$$

So, the circle has **center $(3, 6)$** and **radius $R = 6$**.

Step 2: Find the length of the chord

The length of a chord is given by $2\sqrt{R^2 - p^2}$, where p is the perpendicular distance from the center of the circle to the line $x - y - 1 = 0$.

1. **Calculate the distance p from $(3, 6)$ to $x - y - 1 = 0$:**

$$p = \frac{|(1)(3) + (-1)(6) - 1|}{\sqrt{1^2 + (-1)^2}}$$

$$p = \frac{|3 - 6 - 1|}{\sqrt{2}} = \frac{|-4|}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

2. **Calculate the chord length:**

$$\text{Length} = 2\sqrt{6^2 - (2\sqrt{2})^2}$$

$$\text{Length} = 2\sqrt{36 - 8}$$

$$\text{Length} = 2\sqrt{28} = 2\sqrt{4 \times 7} = 2 \times 2\sqrt{7} = \mathbf{4\sqrt{7}}$$

Question: An ellipse having eccentricity $\frac{1}{\sqrt{3}}$ & equation of its directrix is x = 2√2. A hyperbola whose eccentricity is equal to the length of semi-major axis of ellipse & its length of latus rectum is equal to length of minor axis of ellipse, then the distance between the foci of hyperbola is

Options:

(a) $\frac{16\sqrt{3}}{5}$

(b) $\frac{16\sqrt{6}}{15}$

(c) $\frac{17\sqrt{6}}{6}$

(d) $\frac{15\sqrt{3}}{6}$

Answer: (b)

Solution:

Step 1: Determine the properties of the Ellipse

- **Eccentricity (e_E):** $\frac{1}{\sqrt{3}}$
- **Directrix:** $x = \frac{a_E}{e_E} = 2\sqrt{2}$

Substitute the value of e_E to find the semi-major axis (a_E):

$$\frac{a_E}{1/\sqrt{3}} = 2\sqrt{2} \implies a_E\sqrt{3} = 2\sqrt{2} \implies a_E = \frac{2\sqrt{2}}{\sqrt{3}}$$

Now, find the length of the minor axis ($2b_E$) using the relation $b_E^2 = a_E^2(1 - e_E^2)$:

$$b_E^2 = \left(\frac{8}{3}\right)\left(1 - \frac{1}{3}\right) = \frac{8}{3} \cdot \frac{2}{3} = \frac{16}{9}$$

$$b_E = \frac{4}{3} \implies \text{Minor axis length } (2b_E) = \frac{8}{3}$$

Step 2: Determine the properties of the Hyperbola

- **Eccentricity (e_H):** Given $e_H = a_E = \frac{2\sqrt{2}}{\sqrt{3}}$
- **Latus Rectum (LR_H):** Given $LR_H = \text{Minor axis of ellipse} = \frac{8}{3}$

Using the formula for the latus rectum of a hyperbola ($\frac{2b_H^2}{a_H}$):

$$\frac{2b_H^2}{a_H} = \frac{8}{3} \implies \frac{b_H^2}{a_H} = \frac{4}{3} \implies b_H^2 = \frac{4}{3}a_H$$

Using the eccentricity relation for a hyperbola $b_H^2 = a_H^2(e_H^2 - 1)$:

$$\frac{4}{3}a_H = a_H^2\left(\left(\frac{2\sqrt{2}}{\sqrt{3}}\right)^2 - 1\right)$$

$$\frac{4}{3}a_H = a_H^2\left(\frac{8}{3}-1\right) \Longrightarrow \frac{4}{3}a_H = \frac{5}{3}a_H^2$$

$$4 = 5a_H \Longrightarrow a_H = \frac{4}{5}$$

Step 3: Calculate the distance between the foci

The distance between the foci of a hyperbola is given by $2a_He_H$:

$$\text{Distance} = 2\left(\frac{4}{5}\right)\left(\frac{2\sqrt{2}}{\sqrt{3}}\right) = \frac{16\sqrt{2}}{5\sqrt{3}}$$

To match the options, we rationalize the denominator by multiplying by $\frac{\sqrt{3}}{\sqrt{3}}$:

$$\text{Distance} = \frac{16\sqrt{6}}{5\cdot 3} = \mathbf{\frac{16\sqrt{6}}{15}}$$

Question: The shortest distance between skew lines. $\frac{x+5}{2} = \frac{y-1}{12} = \frac{z-2}{-5}$ $\frac{x-2}{3} = \frac{y-2}{4} = \frac{z-4}{-1}$ **is**

Options:

(a) $\frac{14}{\sqrt{1107}}$

(b) $\frac{13}{\sqrt{1017}}$

(c) $\frac{21}{\sqrt{1701}}$

(d) $\frac{19}{\sqrt{7101}}$

Answer: (b)

Solution:

Step 1: Extract Points and Direction Vectors

From the given equations of the lines:

- **Line 1 (L_1):** Passes through point $A(-5,1,2)$ and is parallel to vector $\vec{b_1} = 2\hat{i} + 12\hat{j} - 5\hat{k}$.
- **Line 2 (L_2):** Passes through point $B(2,2,4)$ and is parallel to vector $\vec{b_2} = 3\hat{i} + 4\hat{j} - \hat{k}$.

Let $\vec{AB}$ be the vector joining the two points:

$$\vec{AB} = (2-(-5))\hat{i} + (2-1)\hat{j} + (4-2)\hat{k} = 7\hat{i} + 1\hat{j} + 2\hat{k}$$

Step 2: Find the Perpendicular Vector ($\vec{b_1} \times \vec{b_2}$)

We find the cross product of the direction vectors to get a vector perpendicular to both lines:

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 12 & -5 \\ 3 & 4 & -1 \end{vmatrix}$$

$$= \hat{i}(-12-(-20)) - \hat{j}(-2-(-15)) + \hat{k}(8-36)$$

$$= \mathbf{8\hat{i} - 13\hat{j} - 28\hat{k}}$$

Now, calculate its magnitude:

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{8^2 + (-13)^2 + (-28)^2}$$

$$= \sqrt{64 + 169 + 784} = \sqrt{\mathbf{1017}}$$

Step 3: Calculate the Shortest Distance

The formula for the shortest distance d is:

$$d = \frac{|\vec{AB} \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

First, find the dot product:

$$\vec{AB} \cdot (\vec{b}_1 \times \vec{b}_2) = (7)(8) + (1)(-13) + (2)(-28)$$

$$= 56 - 13 - 56 = \mathbf{-13}$$

Finally, substitute the values into the distance formula:

$$d = \left| \frac{-13}{\sqrt{1017}} \right| = \frac{\mathbf{13}}{\sqrt{\mathbf{1017}}}$$

Question: The line x - y = 4 intersect the circle $(x - 4)^2 + (y + 3)^2 = 9$ at point P & Q. There is a point M(α, β) on circle such that MP = MQ. Then the value of $|6\alpha + 8\beta|$ is equal to

Options:

(a) $3\sqrt{2}$

(b) $4\sqrt{3}$

(c) $3\sqrt{3}$

(d) $2\sqrt{3}$

Answer: (a)

Solution:

Step 1: Analyze the Circle and the Line

The given circle is $(x-4)^2+(y+3)^2=9$.

- **Center C:** $(4,-3)$
- **Radius R:** $\sqrt{9}=3$

The line intersecting the circle is $x-y=4$.

Step 2: Understand the Geometry of Point M

Point $M(\alpha,\beta)$ is on the circle and is equidistant from P and Q ($MP=MQ$). This means:

1. M must lie on the **perpendicular bisector** of the chord PQ.
2. In any circle, the perpendicular bisector of a chord always passes through the **center** of the circle.

Find the equation of the perpendicular bisector:

- The slope of the chord PQ ($x-y=4$) is $m_1=1$.
- The slope of the perpendicular line is $m_2=-1/1=-1$.
- Since it passes through the center $C(4,-3)$, its equation is:

$$y-(-3)=-1(x-4)$$

$$y+3=-x+4$$

$$\boldsymbol{x+y=1}$$

Since $M(\alpha,\beta)$ lies on this line, we have the relation: $\boldsymbol{\alpha+\beta=1}$.

Step 3: Find the Coordinates of M

Substitute $\beta=1-\alpha$ into the equation of the circle:

$$(\alpha-4)^2+((1-\alpha)+3)^2=9$$

$$(\alpha-4)^2+(4-\alpha)^2=9$$

$$2(\alpha-4)^2=9 \implies (\alpha-4)^2=\frac{9}{2}$$

$$\alpha-4=\pm\frac{3}{\sqrt{2}} \implies \alpha=4\pm\frac{3}{\sqrt{2}}$$

Now, find corresponding β:

$$\beta=1-\alpha=1-\left(4\pm\frac{3}{\sqrt{2}}\right)=-3\mp\frac{3}{\sqrt{2}}$$

Step 4: Calculate $|6\alpha+8\beta|$

Using the relation $\alpha+\beta=1$, we can simplify the expression:

$$6\alpha+8\beta=6(\alpha+\beta)+2\beta=6(1)+2\beta=6+2\beta$$

Now substitute the values of β:

- If $\beta = -3 - \frac{3}{\sqrt{2}}$: $6 + 2\left(-3 - \frac{3}{\sqrt{2}}\right) = 6 - 6 - \frac{6}{\sqrt{2}} = -3\sqrt{2}$
- If $\beta = -3 + \frac{3}{\sqrt{2}}$: $6 + 2\left(-3 + \frac{3}{\sqrt{2}}\right) = 6 - 6 + \frac{6}{\sqrt{2}} = 3\sqrt{2}$

In both cases, the absolute value is:

$$|6\alpha + 8\beta| = 3\sqrt{2}$$

Question: If cos 3θ + 2cos2θ = -2, then sum of all possible solutions in [0, 2π] is

Options:

(a) π

(b) 2π

(c) 3π

(d) 4π

Answer: (d)

Solution:

Step 1: Simplify the equation using identities

Recall the multiple-angle formulas:

- $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$
- $\cos 2\theta = 2\cos^2\theta - 1$

Substitute these into the given equation:

$$(4\cos^3\theta - 3\cos\theta) + 2(2\cos^2\theta - 1) = -2$$

$$4\cos^3\theta + 4\cos^2\theta - 3\cos\theta - 2 = -2$$

$$4\cos^3\theta + 4\cos^2\theta - 3\cos\theta = 0$$

Step 2: Factor and solve for $\cos\theta$

Factor out $\cos\theta$ from the equation:

$$\cos\theta(4\cos^2\theta + 4\cos\theta - 3) = 0$$

This gives us two cases to solve:

Case 1: $\cos\theta = 0$

In the interval $[0, 2\pi]$, the solutions are:

- $\theta = \frac{\pi}{2}$
- $\theta = \frac{3\pi}{2}$

Case 2: $4\cos^2\theta + 4\cos\theta - 3 = 0$

Let $x = \cos\theta$. We solve the quadratic equation $4x^2 + 4x - 3 = 0$:

$$(2x - 1)(2x + 3) = 0$$

So, $x = \frac{1}{2}$ or $x = -\frac{3}{2}$.

Since the value of $\cos\theta$ must be between -1 and 1, we discard $x = -\frac{3}{2}$.

For $\cos\theta = \frac{1}{2}$ in the interval $[0, 2\pi]$, the solutions are:

- $\theta = \frac{\pi}{3}$
- $\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

Step 3: Calculate the sum of all solutions

The possible values for θ are $\{\frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3}\}$.

$$\text{Sum} = \left(\frac{\pi}{2} + \frac{3\pi}{2}\right) + \left(\frac{\pi}{3} + \frac{5\pi}{3}\right)$$

$$\text{Sum} = 2\pi + 2\pi$$

$$\text{Sum} = 4\pi$$

Question: Let $\lim_{x\to 2} \frac{\tan(x-2)\left(rx^2 + (p-2)x - 2p\right)}{(x-2)^2} = 5.$ **If both roots of the equation rx^2 - px + q = 0 lie interval (0, 2) then set of values of q is (α, β] then 16(β - α)2 is equal to**

Options:

(a) 2

(b) 1

(c) 4

(d) 8

Answer: (b)

Solution:

Step 1: Evaluate the limit to find *r* and *p*

The given limit is:

$$\lim_{x\to 2} \frac{\tan(x-2)\cdot(rx^2 + (p-2)x - 2p)}{(x-2)^2} = 5$$

We can rewrite this as:

$$\left(\lim_{x\to 2} \frac{\tan(x-2)}{x-2}\right)\cdot\left(\lim_{x\to 2} \frac{rx^2 + (p-2)x - 2p}{x-2}\right) = 5$$

1. Since $\lim_{\theta\to 0} \frac{\tan\theta}{\theta} = 1$, the first part of the limit is 1.
2. The remaining part must equal 5: $\lim_{x\to 2} \frac{rx^2 + (p-2)x - 2p}{x-2} = 5$.
3. For this limit to be finite, the numerator must be zero at $x = 2$:

$$r(2^2) + (p-2)(2) - 2p = 0 \Longrightarrow 4r + 2p - 4 - 2p = 0 \Longrightarrow 4r = 4 \Longrightarrow \mathbf{r = 1}$$

4. Substitute $r = 1$ back into the limit:

$$\lim_{x\to 2}\frac{x^2+(p-2)x-2p}{x-2} = \lim_{x\to 2}\frac{(x-2)(x+p)}{x-2} = \lim_{x\to 2}(x+p) = 5$$

$$2+p=5 \Longrightarrow \mathbf{p=3}$$

Step 2: Find the set of values for q

Now we consider the quadratic equation $rx^2 - px + q = 0$. Substituting $r = 1$ and $p = 3$:

$$x^2 - 3x + q = 0$$

For both roots to lie in the interval **(0, 2)**, three conditions must be met:

1. **Discriminant ($D \geq 0$):**

$$(-3)^2 - 4(1)(q) \geq 0 \Longrightarrow 9 - 4q \geq 0 \Longrightarrow \mathbf{q \leq \frac{9}{4}}$$

2. **Function value at the boundaries ($f(0) > 0$ and $f(2) > 0$):**

   1. $f(0) = 0 - 0 + q > 0 \Longrightarrow \mathbf{q > 0}$
   2. $f(2) = 2^2 - 3(2) + q > 0 \Longrightarrow 4 - 6 + q > 0 \Longrightarrow \mathbf{q > 2}$

3. **Vertex location ($0 < -\frac{b}{2a} < 2$):**

$$0 < \frac{3}{2} < 2 \quad \text{(This is already true.)}$$

Combining these conditions, we find that $2 < q \leq \frac{9}{4}$. Thus, the set is $(2, 2.25]$, which gives us:

$$\alpha = 2 \quad \text{and} \quad \beta = \frac{9}{4}$$

Step 3: Calculate the final expression

Now, substitute the values of α and β into the required expression:

$$\beta - \alpha = \frac{9}{4} - 2 = \frac{1}{4}$$

$$(\beta-\alpha)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$16(\beta-\alpha)^2 = 16 \times \frac{1}{16} = \mathbf{1}$$

Question: If the mean and variance of the observations 2, 4, α, 8, β, 12, 14 (where $\alpha < \beta$) are 8 and 16 respectively. Then, the equation whose roots are $3\alpha + 2$ and $4\beta + 1$ is

Options:

(a) $x^2 - 16x + 820 = 0$

(b) $x^2 - 60x + 340 = 0$

(c) $x^2 - 60x + 81 - 0$

(d) $x^2 - 61x + 810 = 0$

Answer: (a)

Solution:

Step 1: Use the Mean to find $\alpha + \beta$

The mean $(\bar{x})$ of the 7 observations is given as 8:

$$\bar{x} = \frac{2+4+\alpha+8+\beta+12+14}{7} = 8$$

$$40 + \alpha + \beta = 56$$

$$\boldsymbol{\alpha + \beta = 16} \quad \ldots\text{(Equation 1)}$$

Step 2: Use the Variance to find $\alpha^2 + \beta^2$

The variance (σ^2) is given as 16:

$$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2 = 16$$

$$\frac{2^2+4^2+\alpha^2+8^2+\beta^2+12^2+14^2}{7} - 8^2 = 16$$

$$\frac{4+16+\alpha^2+64+\beta^2+144+196}{7} = 16 + 64$$

$$\frac{424+\alpha^2+\beta^2}{7} = 80$$

$$424 + \alpha^2 + \beta^2 = 560$$

$$\boldsymbol{\alpha^2 + \beta^2 = 136} \quad \ldots\text{(Equation 2)}$$

Step 3: Solve for α and β

Using the identity $(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$:

$$16^2 = 136 + 2\alpha\beta$$

$$256 = 136 + 2\alpha\beta \implies 2\alpha\beta = 120 \implies \boldsymbol{\alpha\beta = 60}$$

We now have two numbers that sum to 16 and multiply to 60. These are the roots of $t^2 - 16t + 60 = 0$, which factors as $(t-6)(t-10) = 0$.
Given the condition $\alpha < \beta$, we have:
$\boldsymbol{\alpha = 6}$ and $\boldsymbol{\beta = 10}$.

Step 4: Form the new quadratic equation

The roots of the required equation are:

- **Root 1 (r_1):** $3\alpha + 2 = 3(6) + 2 = \mathbf{20}$
- **Root 2 (r_2):** $4\beta + 1 = 4(10) + 1 = \mathbf{41}$

Now, find the sum and product of these roots:

- **Sum of roots (S):** $20 + 41 = \mathbf{61}$
- **Product of roots (P):** $20 \times 41 = \mathbf{820}$

The quadratic equation is given by $x^2 - Sx + P = 0$:

$$x^2 - 61x + 820 = 0$$